Neutronics Modeling of the Tritium Breeding LIBRA Experiment

Collin S. Dunn, Ethan E. Peterson, Kevin B. Woller, Sara E. Ferry, Stefano Segantin, Rémi Delaporte-Mathurin, Zachary Hartwig, Dennis G. Whyte

MIT Plasma Science and Fusion Center

PSFC

1. Objectives:

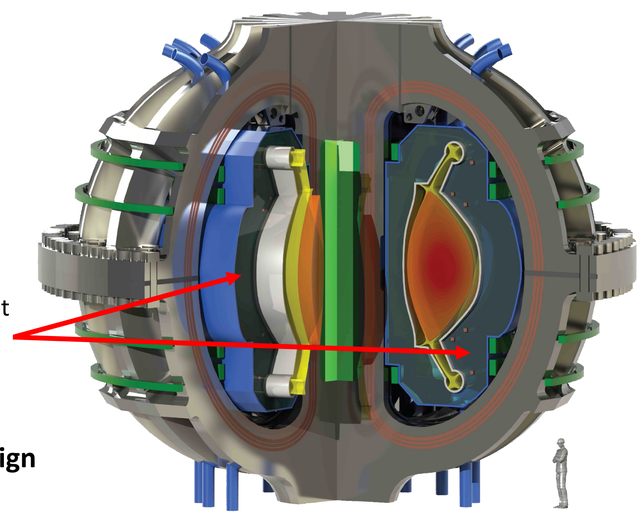
Design shielding for the LIBRA experiment with the following requirements:

- ☐ Reduce combined neutron and photon dose to <2 mrem/hr for radiation workers
- ☐ Reduce combined neutron and photon dose to the general public to <0.1 mrem/hr
- ☐ Minimize shielding mass to <26 metric tons
- ☐ Ensure shielding is modular and can be constructed without the use of cranes

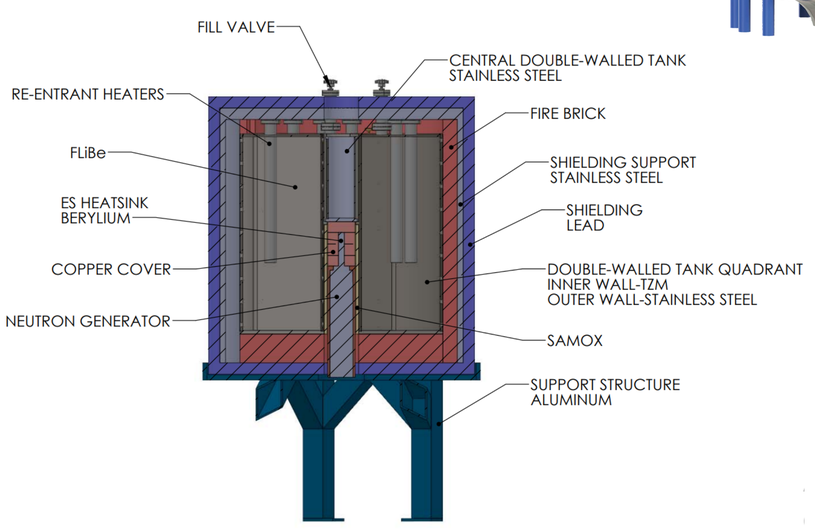
2. Background:

- LIBRA: Liquid Immersion Blanket Robust Accountancy
- The liquid immersion blanket is a simple tritium-breeding concept capable of achieving high tritium-breeding ratios (TBR) that has been proposed for fusion experiments such as the ARC device

ARC Fusion Reactor Concept



LIBRA Experimental Prototype Design

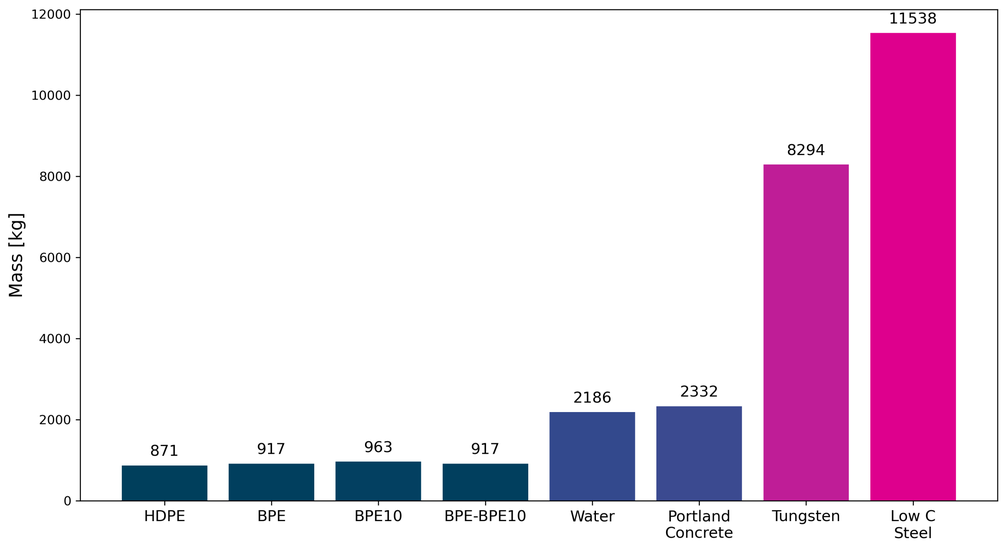


- The LIBRA experiment will be the first molten-salt, large-scale tritium-breeding experiment using FLiBe ($2LiF\text{-}BeF_2$) [Ferry 2022]
  - 1000 kg of FLiBe
  - 10^{10} neutrons/second DT neutron generator (14.1 MeV neutrons)

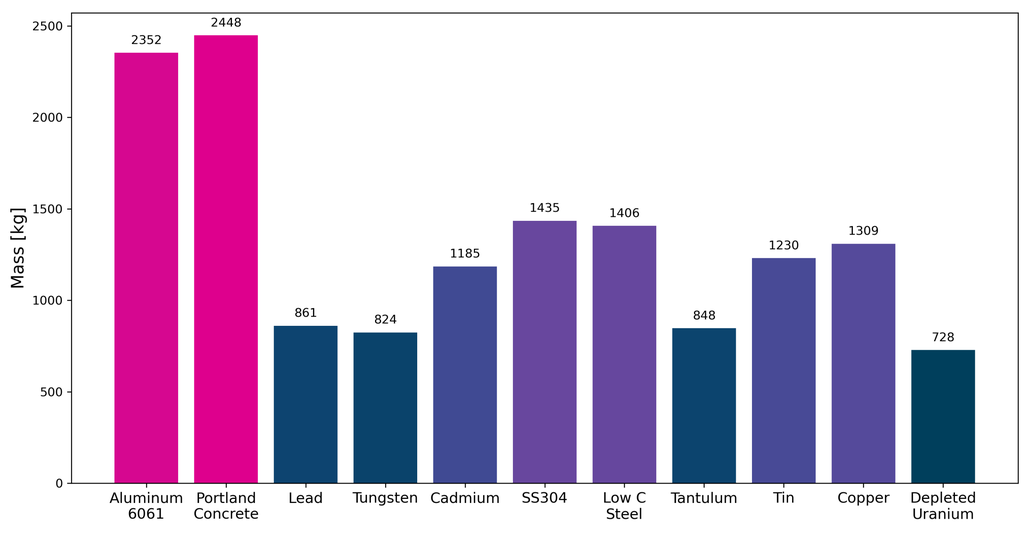
3. Analyze materials for highest attenuation density

- **LIBRA has major mass restrictions due to limitations of the laboratory structure, so shielding materials selected should have the highest neutron/photon attenuation per mass and be commercially available**
- Materials were modeled using a spherical shell structure in OpenMC with a point source located in the center of the spherical shell
- The neutron point source released 14.1 MeV monoenergetic neutrons
- The photon point source released photons with an energy distribution tallied on the LIBRA experiment
- The materials were evaluated based on the mass required to attenuate the neutron dose and photon dose by 99% as compared to the vacuum dose values.
- 5 wt.% polyethylene (BPE) and high density polyethylene (HDPE) produced the highest neutron attenuation densities
- Lead, tungsten, tantalum, and depleted uranium produced the highest photon attenuation densities, but lead and tungsten are typically more commercially available and cheaper than tantalum or depleted uranium (Becker 2018)

99% Neutron Dose Attenuation from 14.1 MeV Incident Neutrons

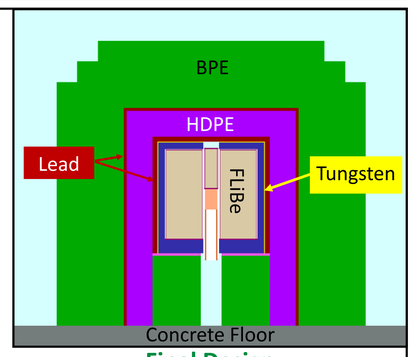


99% Secondary Photon Dose Attenuation from LIBRA Photons



4. Model and optimize shielding design

- Materials with the highest attenuations densities were simulated in OpenMC in layered cylindrical configurations around the LIBRA experiment by placing the photon shield on the inside, outside, or in various layers
- Since the dose contours are more spherically shaped, mass reduction achieved by cutting out stepped-cylindrical regions in outer shield



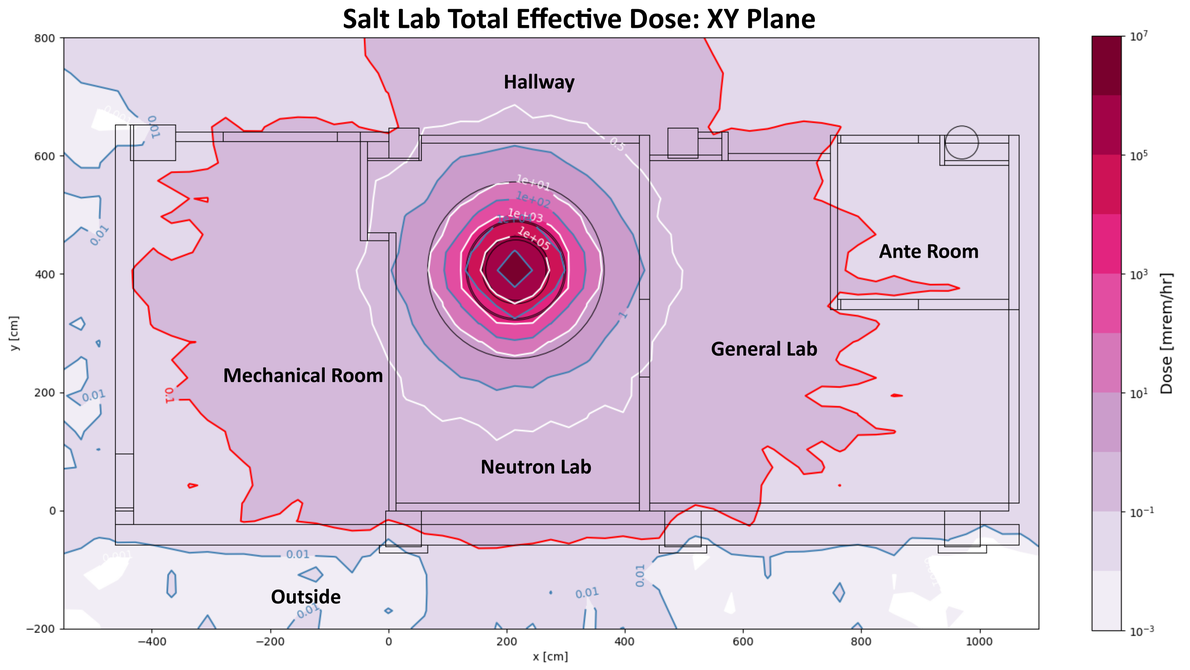
Final Design

Shielding Description	Effective Dose at 175 cm [mrem/hr]	Shielding Mass [metric tons]
Inner: 5 cm lead Outer: 90 cm BPE	0.88 ± 0.02	20.9
Inner: 90 cm BPE Outer: 5 cm lead	0.65 ± 0.03	36.2
Inner P-shield: 5 cm lead **Inner N-shield: 30cm HDPE** **Middle P-shield: 2.54 cm lead** **Outer N-shield: 65 cm BPE**	**0.64 ± 0.04**	**22.8**

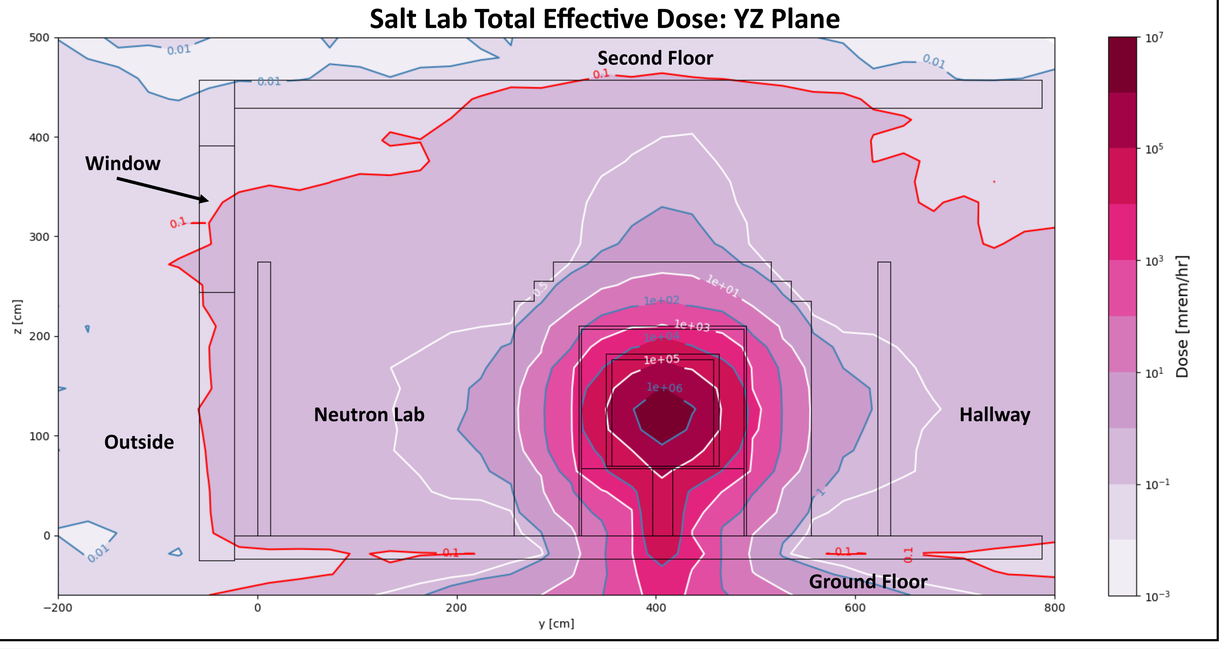
5. Simulate optimized design in laboratory model

- OpenMC model created of the salt lab using proposed floor plans and the shielding design for the LIBRA experiment
- Only first floor modeled, so effects due to basement, outside soil, and second floor are neglected
- **Maximum dose rate to radiation workers: 0.7 mrem/hr**
- **Maximum dose rate to public: 0.7 mrem/hr**

Salt Lab Total Effective Dose: XY Plane



Salt Lab Total Effective Dose: YZ Plane



6. Conclusion

By using the lead-HDPE-lead-BPE cylindrical configuration:

- ☑ **Radiation worker dose limited to <0.7 mrem/hr**
- ☑ **Experiment mass is 24.4 metric tons**
- ☑ **HDPE, BPE, and lead bricks can be installed modularly**
- ☐ **Public dose NOT limited to <0.1 mrem/hr**

- Additional shielding around the experiment or within the lab walls will be needed
- The salt lab floor plans should be adjusted to accommodate for the additional space needed by the shielding
- Further analysis using more detailed models of the LIBRA experiment and the salt lab should be carried out
- Shutdown dose due to neutron activation was not studied, but will be important for maintenance operations
- This analysis demonstrates that medium to large scale liquid-immersion-blanket experiments can be conducted safely with regard to radiation exposure in a university environment.

7. References

Becker, Leon. 2018. “Chemical Elements by Market Price.” October 9, 2018. http://www.leonland.de/elements_by_price/en/.

Detwiler, Rebecca S., Ronald J. McConn, Thomas F. Grimes, Scott A. Upton, and Eric J. Engel. 2021. “Compendium of Material Composition Data for Radiation Transport Modeling.” PNNL-15870-Revision-2. Pacific Northwest National Lab. (PNNL), Richland, WA (United States). https://doi.org/10.2172/1782721.

Ferry, Sara E., Kevin B. Woller, Ethan E. Peterson, Caroline Sorensen, and Dennis G. Whyte. 2023. “The LIBRA Experiment: Investigating Robust Tritium Accountancy in Molten FLiBe Exposed to a D-T Fusion Neutron Spectrum.” *Fusion Science and Technology* 79 (1): 13–35. https://doi.org/10.1080/15361055.2022.2078136.

Williams, Ernest Jonathan. 2015. “Development of a Neutron Generator Facility at Simon Fraser University.” Simon Fraser University. April 27, 2015. https://summit.sfu.ca/item/15323.